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Regulation Landscape

CEC 300: Computing in Aviation and Aerospace

Authors: Richard S. Stansbury

Terminology

Airworthiness Certificate:

- a certificate stating an aircraft has been deemed airworthy, which includes conformance to its type certificate and continued airworthiness (maintenance, repair, etc.)
- **Standard**
- **Special:** primary, restricted, limited, light-sport, and experimental (i.e., R&D, crew training, exhibition, marketing)

Type Certificate (TC): a certificate acknowledging airworthiness compliance for an aircraft type.

Production Certificate: a certificate authorizing production of aircraft based upon a type certificate.

Special Conditions: can be designated by the FAA for an aircraft type that does not conform to existing airworthiness standards of 14 CFR because of novel / unique design features.

Terminology

Airworthiness Directive (AD):

- "Airworthiness Directives (ADs) are legally enforceable rules issued by the FAA in accordance with 14 CFR part 39 to correct an unsafe condition in a product. 14 CFR part 39 defines a product as an aircraft, aircraft engine, propeller, or appliance." (FAA)
- Responsibility of aircraft owner / operator to take action

Advisory Circular (AC):

- FAA published documents that provide guidance regarding acceptable means of compliance for FAA standards such as airworthiness
- Not binding and not the sole means of compliance
- Sometimes reference industry-base standards (RTCA, ASTM, etc.)

Technical Standard Order (TSO):

- "A TSO is a minimum performance standard for specified materials, parts, and appliances used on civil aircraft. When authorized to manufacture a material, part, or appliances to a TSO standard, this is referred to as TSO authorization. Receiving a TSO authorization is both design and production approval."
- A manufacturer can obtain a TSO authorization for a product indicating that their product meets the standards and is authorized to produce the product

Orders:

- Documents defining FAA policy and procedures for FAA personnel and representatives
- Sometimes the orders can provide guidance to developers, operators, etc. on how regulation is interpreted and compliance is assessed.

From CAB to FAA

- 1938 Civil Aeronautics Board Created
  - Civil Aeronautics Act (52 Statute 973)
- 1958 Becomes the Federal Aviation Agency
  - CARs become FARs
- 1966 Federal Aviation Administration

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FAA Regulations, Title 14 Code of Federal Regulations

Title	Volume	Chapter	Browse Parts	Regulatory Entity
Title 14 Aeronautics and Space	1	I	1-59	FEDERAL AVIATION ADMINISTRATION, DEPARTMENT OF TRANSPORTATION
	2		60-109	
	3		110-199	
	4	II	200-399	OFFICE OF THE SECRETARY, DEPARTMENT OF TRANSPORTATION (AVIATION PROCEEDINGS)
		III	400-1199	COMMERCIAL SPACE TRANSPORTATION, FEDERAL AVIATION ADMINISTRATION, DEPARTMENT OF TRANSPORTATION
	5	V	1200-1299	NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
		VI	1300-1399	AIR TRANSPORTATION SYSTEM STABILIZATION

Under public statutory law, the authority for lawmaking with respect to the regulation of aviation was granted to the FAA so that it can write regulatory laws with defined enforcement procedures (fines, loss of certificates, etc.)

Generating new laws can be cumbersome.
General Process:

- Aviation rulemaking committees (ARCs) can assist in developing new rules
- A draft rule must be vetted by FAA and DOT
- A notice of public rulemaking is issued
- A period of public comment occurs
- FAA must address all comments
- Revision is made toward final regulation
- Approval through FAA and DOT before becoming a new regulatory law.

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Title 14 CFR Part 21

Foundational Standard for Airworthiness Requirement

Aircraft designers seek a **Type Certificate (TC)** certifying the aircraft design is airworthy.

- Entire aircraft certified under a TC
- Onboard equipment cannot be modified under a type certificate.
- Does not certify the individual components.

Supplementary type certificates (STCs) can be issued for aircraft that are modifications to an existing TC'd aircraft.

- Approves airworthiness
- Operational limits of the modified aircraft

Airworthiness Certification determined based upon the safety case put forward by design engineers. Must meet hardware and software Design Assurance Levels.

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Delegates the airworthiness certification requirements of airworthiness to aircraft and operational type via additional Parts of Title 14 CFR:

- 14 CFR Part 23 (general aviation),
- 14 CFR Part 25 (transport category),
- 14 CFR Part 27 (general aviation rotorcraft), and
- 14 CFR Part 29 (transport category).

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Snippets from 14 CFR Part 23

Subpart A—General

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§23.2000 Applicability and definitions.

(a) This part prescribes airworthiness standards for the issuance of type certificates, and changes to those certificates, for airplanes in the normal category.

(b) For the purposes of this part, the following definition applies:

Continued safe flight and landing means an airplane is capable of continued controlled flight and landing, possibly using emergency procedures, without requiring exceptional pilot skill or strength. Upon landing, some airplane damage may occur as a result of a failure condition.

§23.2105 Performance data.

(a) Unless otherwise prescribed, an airplane must meet the performance requirements of this subpart in—

(1) Still air and standard atmospheric conditions at sea level for all airplanes; and

(2) Ambient atmospheric conditions within the operating envelope for levels 1 and 2 high-speed and levels 3 and 4 airplanes.

(b) Unless otherwise prescribed, the applicant must develop the performance data required by this subpart for the following conditions:

(1) Airport altitudes from sea level to 10,000 feet (3,048 meters); and

(2) Temperatures above and below standard day temperature that are within the range of operating limitations, if those temperatures could have a negative effect on performance.

(c) The procedures used for determining takeoff and landing distances must be executable consistently by pilots of average skill in atmospheric conditions expected to be encountered in service.

(d) Performance data determined in accordance with paragraph (b) of this section must account for losses due to atmospheric conditions, cooling needs, and other demands on power sources.

Each part is divided into:
Subparts: a grouping of related sections.
Sections: a grouping of paragraphs that define a rule, definition, or guidance for regulation
Paragraphs and sub-paragraphs: provide specific details

14 CFR § 23.2105(b)
Interpret as
Title 14 of Code of Federal Reg.
Part 23
Section 2105
Paragraph b
§§ is used if discussing multiple sections.

Note: Part 23 was updated in 2017, which included a significant rework to be less prescriptive with design requirements and focus upon performance-based airworthiness standards.
i.e. what must be achieved, but not how.

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TSOs

- Technical Standard Orders (TSOs) are issued for standard components that are interchangeable such as altimeters.
- The performance standards are defined in the TSO and the equipment is marked with a TSO number.

Department of Transportation
Federal Aviation Administration
Aircraft Certification Service
Washington, DC

TSO-C10b

Date: 9/1/59

Technical Standard Order

Subject: TSO-C10b, ALTIMETER, PRESSURE ACTUATED, SENSITIVE TYPE

Part 514 contains minimum performance standards and specifications of materials parts, processes, and appliances used in aircraft and implements the provisions of sections 3.18, 4a.31, 4b.18, 6.18 and 7.18 of the Civil Air Regulations. The regulation uses the Technical Standard

SAE standard for Altimeters

- TSO shows FAA acceptance
  Later, CARs updated with section 514.20.

SAE Aerospace — An SAE International Group

AERONAUTICAL STANDARD **AS 392 C**

Issued 1947-03
Revised 1959-02
Reaffirmed 1993-05
Rev Nov 2008-02

ALTIMETER, PRESSURE ACTUATED SENSITIVE TYPE

1. PURPOSE: To specify minimum requirements for Pressure Actuated Sensitive Altimeters for use in aircraft, the operation of which may subject the instrument to the environmental conditions specified in paragraph 3.3.

2. SCOPE: This Aeronautical Standard covers two (2) basic types of instruments as follows:

TYPE I - Range 35,000 feet. Barometric Pressure. Scale range at least 28.1 - 30.99 inches of mercury (946-1049 millibars). May include markers working in conjunction with the Barometric Pressure Scale to indicate pressure altitude.

TYPE II- Range 50,000 feet. Barometric Pressure. Scale range at least 28.1 - 30.99 inches of mercury (946-1049 millibars). May include markers working in conjunction with the Barometric Pressure Scale to indicate pressure altitude.

3. GENERAL REQUIREMENTS:

3.1 Materials and Workmanship:

3.1.1 Materials: Materials shall be of a quality which experience and/or tests have demonstrated to be suitable and dependable for use in aircraft instruments.

3.1.2 Workmanship: Workmanship shall be consistent with high-grade aircraft instrument manufacturing practice.

3.2 Identifications: The following information shall be legibly and permanently marked on the instrument or attached thereto:

a. Name of instrument (Altimeter)
b. Aeronautical Standard AS-392C
c. Manufacturer's part number
d. Manufacturer's Serial number or date of manufacture
e. Manufacturer's name and/or trademark
f. Range

3.3 Environmental Conditions: The following conditions have been established as design requirements only. Tests shall be conducted as specified in Sections 5, 6 and 7.

3.3.1 Temperature: When installed in accordance with the instrument manufacturer's instructions, the instrument shall function over the range of ambient temperature of -30C to 50C and shall not be adversely affected by exposure to temperature of -65C to 70C.

3.3.2 Humidity: The instruments shall function and shall not be adversely affected when exposed to any relative humidity in the range from 0 to 95% at a temperature of approximately 32C.

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Fax: 724-776-0790
Email: CustomerService@sae.org
http://www.sae.org
SAE WEB ADDRESS:

§ 514.20 Aircraft altimeter, pressure actuated, sensitive type—TSO–C10b.

(a) *Applicability*—(1) *Minimum performance standards.* Minimum performance standards are hereby established for aircraft altimeters which specifically are required to be approved for use on civil aircraft of the United States. New models of altimeters manufactured for such use on or after September 1, 1959, shall meet the standards set forth in SAE Aeronautical Standard AS 392C,[1] "Altimeter, Pressure Actuated Sensitive Type," revision date February 1, 1959,[2] with the exceptions listed in subparagraph (2) of this paragraph. Altimeters approved under prior issuances of this section may continue to be manufactured under the earlier provisions.

(2) *Exceptions.* (i) The following specifically numbered paragraphs in AS 392C do not concern minimum performance and therefore are not essential to compliance with this section: 3.1, 3.1.1, 3.1.2, 3.2, 3.2(a)(b)(c)(d)(e)(f).

(ii) In lieu of section 7, in AS 392C, it is a requirement that the altimeters covered by this section be capable of successfully passing the tests in paragraphs 7.1 through 7.5 and an External Case Pressure Test which is as follows:

External Case Pressure Test. The static pressure source of the instrument shall be sealed when an ambient temperature of 25°C and an ambient pressure of 29.92 inches (absolute) of mercury have been achieved. The ambient pressure shall then be increased at a rate of 20 inches of mercury in two seconds to 50 inches (absolute) of mercury and held at that pressure for three minutes. There shall be no adverse effect on the instrument or its accuracy.

(iii) The "Reference Section" under Table II of AS 392C is not applicable.

(b) *Marking.* In lieu of the weight specified in paragraph (c) of § 514.3, the range shall be shown.

(c) *Data requirements.* One copy each of the following shall be furnished to the Chief, Engineering and Manufacturing Division, Federal Aviation Agency, Washington 25, D.C.:

Airworthiness Directives

Legally enforceable directives issued by the FAA to correct an unsafe condition in a product.

- Identifies the issue
- Defines the required procedure to address the issue
- Applicable toward aircraft, aircraft engines, propellers, or appliances

- Types include:
  - Notice of Public Rulemaking (NPRM),
  - Final Rule – Public Comment, and
  - Emergency NPRM

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Battery Damage in the 787

Resulted in 2013 grounding of the Boeing 787

FAA
Aviation Safety

EMERGENCY AIRWORTHINESS DIRECTIVE

www.faa.gov/aircraft/safety/alerts/

DATE: January 16, 2013
AD #: 2013-02-51

Emergency airworthiness directive (AD) 2013-02-51 is sent to owners and operators of The Boeing Company Model 787-8 airplanes.

Background

This emergency AD was prompted by recent incidents involving lithium ion battery failures that resulted in release of flammable electrolytes, heat damage, and smoke on two Model 787-8 airplanes. The cause of these failures is currently under investigation. These conditions, if not corrected, could result in damage to critical systems and structures, and the potential for fire in the electrical compartment.

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2013-02-51 The Boeing Company: Directorate Identifier 2013-NM-016-AD.

(a) Effective Date

This Emergency AD is effective upon receipt.

(b) Affected ADs

None.

(c) Applicability

This AD applies to all The Boeing Company Model 787-8 airplanes, certificated in any category.

(d) Subject

Joint Aircraft System Component (JASC)/Air Transport Association (ATA) of America Code 24, Electrical power.

(e) Unsafe Condition

This AD was prompted by recent incidents involving lithium ion battery failures that resulted in release of flammable electrolytes, heat damage, and smoke on two Model 787-8 airplanes. The cause of these failures is currently under investigation. We are issuing this AD to prevent damage to critical systems and structures, and the potential for fire in the electrical compartment.

(f) Compliance

Comply with this AD within the compliance times specified, unless already done.

(g) Modification or Other Action

Before further flight, modify the battery system, or take other actions, in accordance with a method approved by the Manager, Seattle Aircraft Certification Office (ACO), FAA.

(h) Alternative Methods of Compliance (AMOCs)

(1) The Manager, Seattle ACO, FAA, has the authority to approve AMOCs for this AD, if requested using the procedures found in 14 CFR 39.19. In accordance with 14 CFR 39.19, send your request to your principal inspector or local Flight Standards District Office, as appropriate. If sending information directly to the manager of the ACO, send it to the attention of the person identified in the Related Information section of this AD. Information may be emailed to: 9-ANM-Seattle-ACO-AMOC-Requests@faa.gov.

(2) Before using any approved AMOC, notify your appropriate principal inspector, or lacking a principal inspector, the manager of the local flight standards district office/certificate holding district office.

Airworthiness directive's sections serve specific purposes:

- When and under what conditions is it applicable.
- Impacted component
- Safety issue
- Corrective action
- AMOC required if following a different approach to address the equipment and unsafe condition.

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Other Agencies

- **ICAO, a United Nations Organization**
- Coordinates aviation standards across the world adopting and harmonizing standards and practices including navigation systems, infrastructure, aircraft identifiers, and many other regulations that need harmonization across borders.

- **Civil Aviation Authorities (CAAs)**
  - CAAs are the analogue to the FAA in other countries or regions throughout the world.
  - Examples:
    - Europe: European Aviation Safety Agency (EASA)
    - Australia: Civilization Aviation Safety Agency (CASA)

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Other Regulatory Bodies

- International Telecommunication Union (ITU):
  - International body that establishes RF frequency allocations across regions.
  - Includes RF communication with satellites orbiting the Earth

- Federal Communication Commission (FCC):
  - Oversees standards for radio transmissions with categories of ground-based fixed and mobile ground transmitters and aeronautical radionavigation signals, and mobile aeronautical transmitters.

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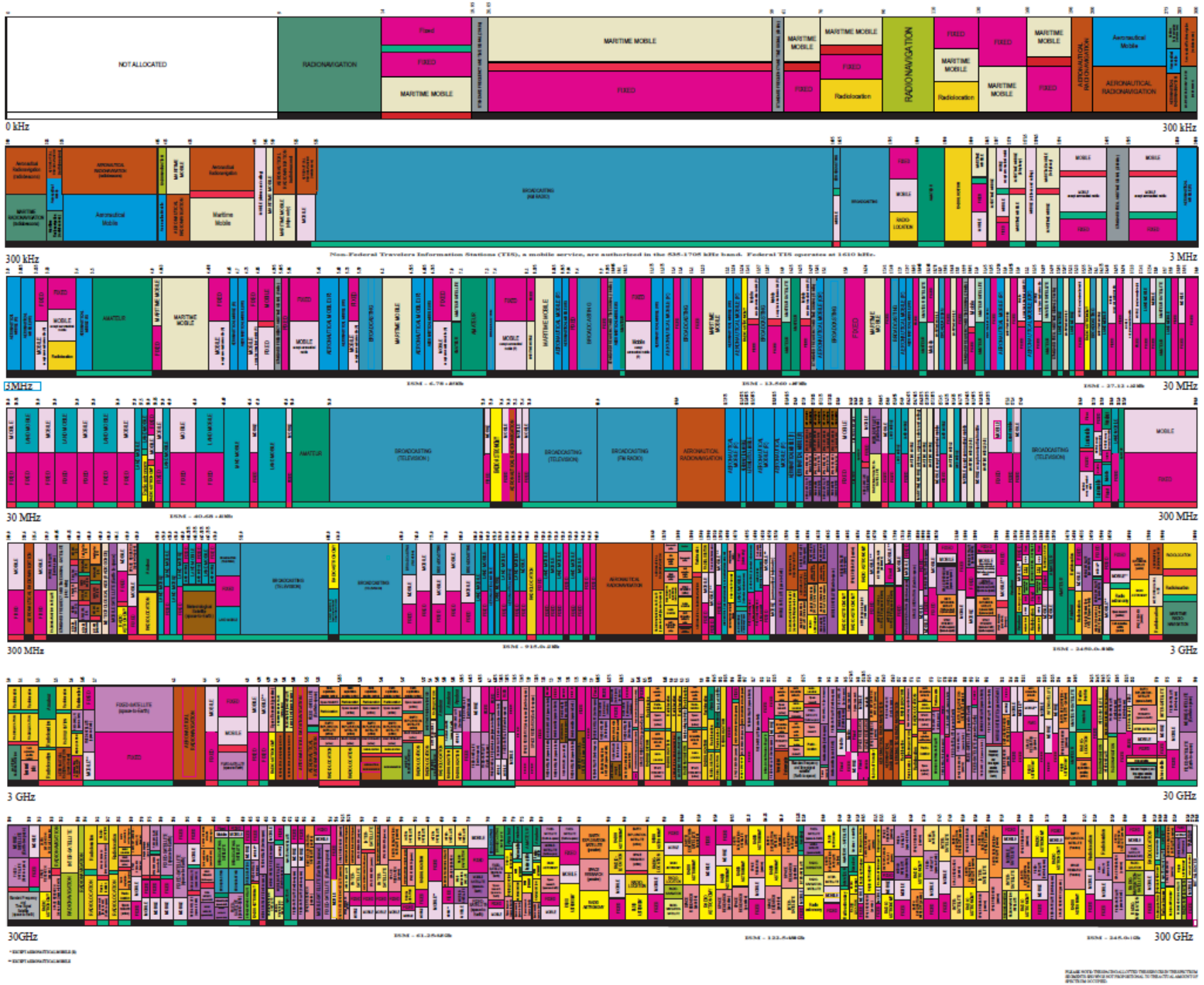
UNITED STATES FREQUENCY ALLOCATIONS

THE RADIO SPECTRUM



RADIO SERVICES COLOR LEGEND

- AERONAUTICAL MOBILE
- AERONAUTICAL MOBILE SATELLITE
- AERONAUTICAL RADIONAVIGATION
- AMATEUR
- AMATEUR SATELLITE
- BROADCASTING
- BROADCASTING SATELLITE
- EARTH EXPLORATION SATELLITE
- FIXED
- FIXED SATELLITE
- INTER-SATELLITE
- LAND MOBILE
- LAND MOBILE SATELLITE
- MARITIME MOBILE
- MARITIME MOBILE SATELLITE
- MARITIME RADIONAVIGATION
- METEOROLOGICAL
- METEOROLOGICAL SATELLITE
- MOBILE
- MOBILE SATELLITE
- RADIO ASTRONOMY
- RADIODETERMINATION SATELLITE
- RADIOLOCATION
- RADIOLOCATION SATELLITE
- RADIONAVIGATION
- RADIONAVIGATION SATELLITE
- SPACE OPERATION
- SPACE RESEARCH
- STANDARD FREQUENCY AND TIME SIGNAL
- STANDARD FREQUENCY AND TIME SIGNAL SATELLITE

ACTIVITY CODE

- FEDERAL EXCLUSIVE
- FEDERAL/NON-FEDERAL SHARED
- NON-FEDERAL EXCLUSIVE

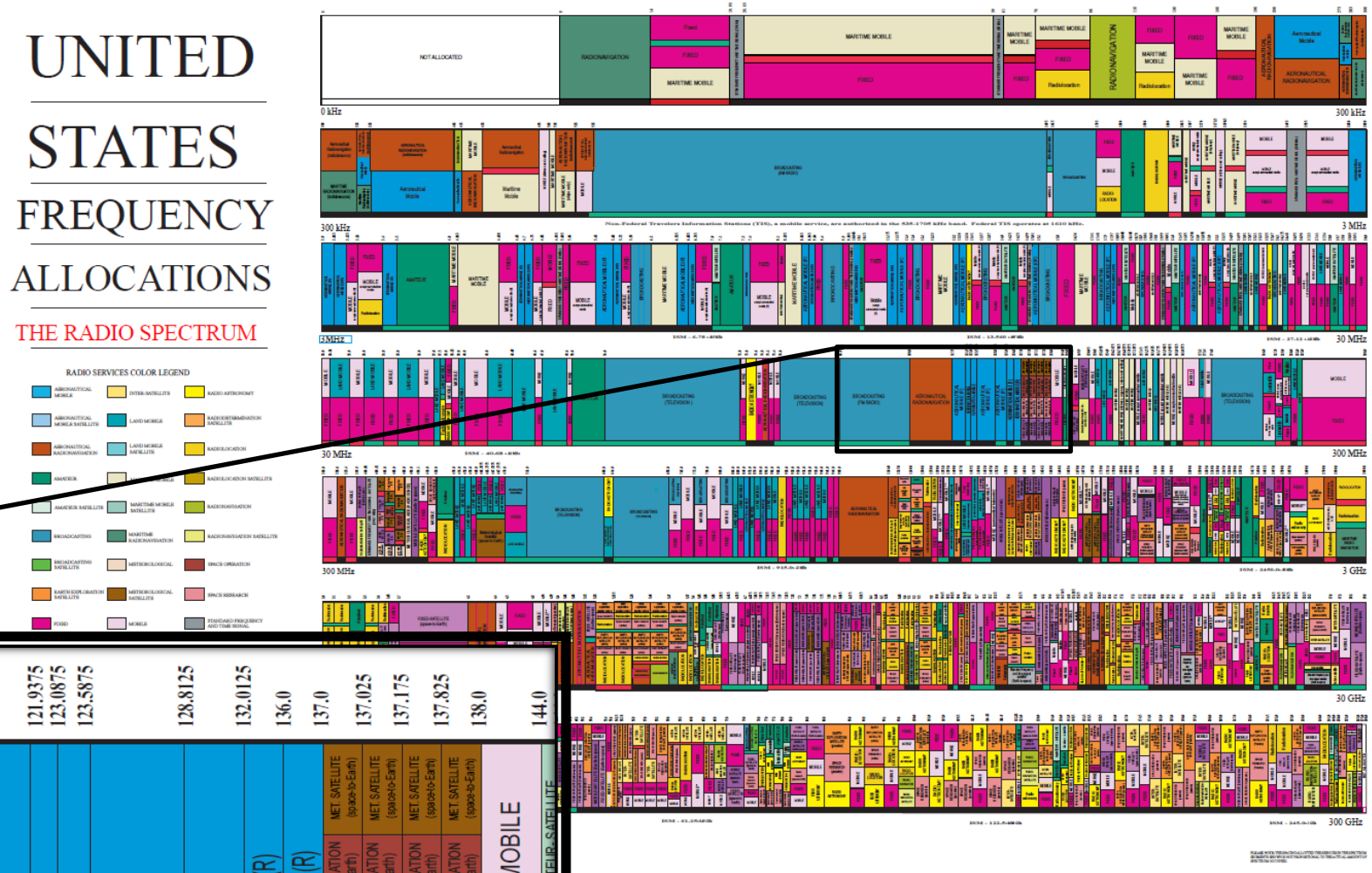
ALLOCATION USAGE DESIGNATION

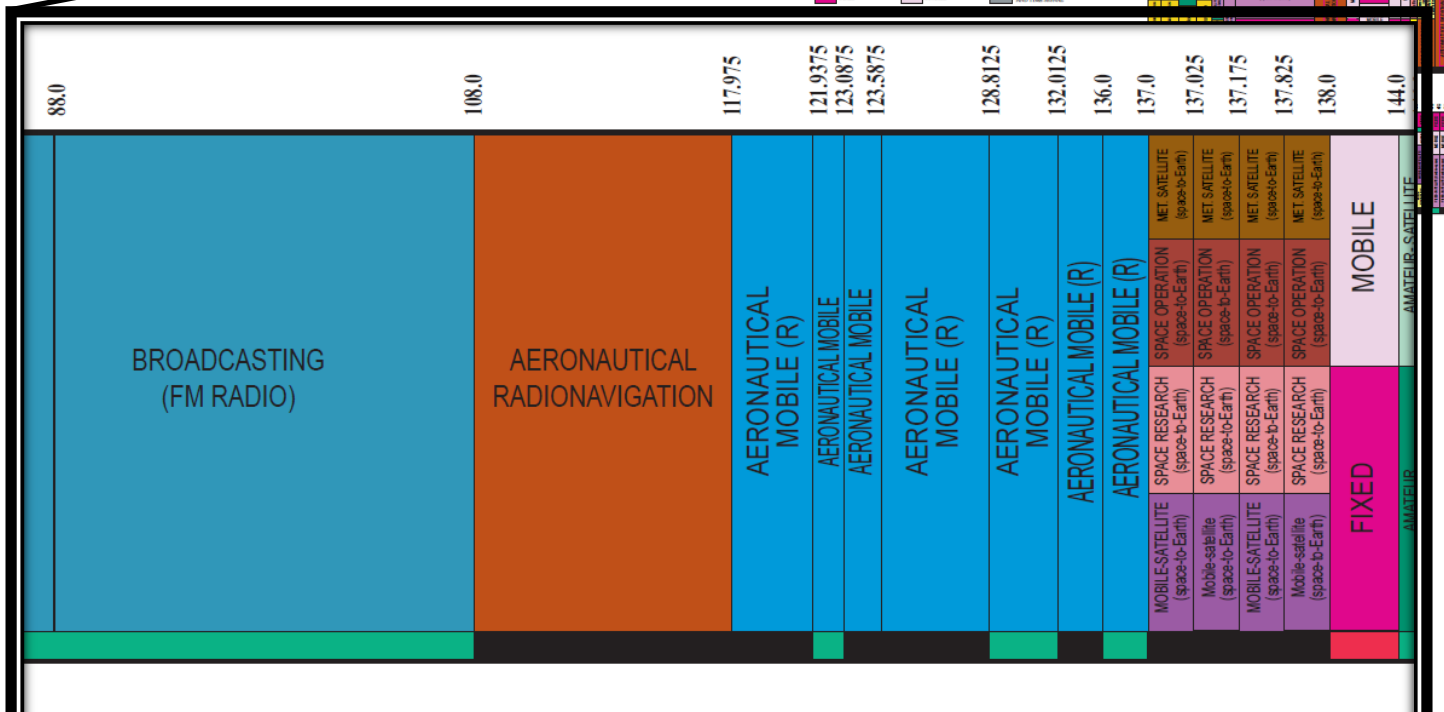
SERVICE	EXAMPLE	DESCRIPTION
Primary	FIXED	Capital Letters
Secondary	Mobile	1st Capital with lower case letters

U.S. DEPARTMENT OF COMMERCE
National Telecommunications and Information Administration
Office of Spectrum Management

JANUARY 2016

Radio Spectrum





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Aviation/Aerospace Standards

- **Standards Development Organizations** produce aviation and aerospace standards that address safety, reliability, and interoperability of aircraft/spacecraft systems.
  - Produce industry-supported consensus standards.
- **RTCA, Inc.** produce a variety of aviation standards used within this industry and relevant to this course.
  - e.g RTCA DO-178c: Airborne Software Certification defines standards to develop software to meet design assurance level based on system criticality and failure risk.

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Other SDOs

- **International Organization for Standards (ISO)** – operations, data, production, operation, and disposal
- **ASTM International** – materials used in spacecraft
- **SAE International** – space technologies, components, and materials
- **IEEE** – wired and wireless communication standards
- **Consultative Committee for Space Data Styems (CCSDS)** – data and communication protocols to ensure interoperability between organizations and nations.
- **American Institute of Aeronautics and Astronautics (AIAA)** – national aerospace standards and support of some ISO technical groups

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MOPS and MASPS

MASPS Minimum Aviation System Performance Standards

From RTCA's website

MASPS specify characteristics that are useful to designers, installers, manufacturers, service providers and users of systems intended for operational use within a defined airspace. Where the systems are global in nature, international applications are taken into consideration (often working with European Organisation for Civil Aviation Equipment (EUROCAE) and/or International Civil Aviation Organization (ICAO)). MASPS describe the system (subsystems / functions) and provide information needed to understand the rationale for system characteristics, operational goals, requirements and typical applications. Definitions and assumptions essential to proper understanding of MASPS are provided as well as minimum system test procedures to verify system performance compliance (e.g., end-to-end performance verification).

MOPS and MASPS

MOPS Minimum Operational Performance Standards

From RTCA's website

MOPS provide standards for specific equipment(s) useful to designers, manufacturers, installers and users of the equipment. The word "equipment" used in MOPS includes all components and units necessary for the system to properly perform its intended function(s). MOPS provide the information needed to understand the rationale for equipment characteristics and requirements stated, describe typical equipment applications and operational goals, and establish the basis for required performance under the standard. Definitions and assumptions essential to proper understanding are provided as well as installed equipment tests and operational performance characteristics for equipment installations.

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MOPS and MASPS

Title:	**Minimum Operational Performance Standards (MOPS) for Helicopter Terrain Awareness and Warning System (HTAWS) Airborne Equipment**
DO #:	**DO-309**

	Minimum Aviation System Performance Standards (MASPS) for Traffic Information Service – Broadcast (TIS-B)
DO #:	**DO-286B**

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Testing

Title:	**Environmental Conditions and Test Procedures for Airborne Equipment**
DO #:	**DO-160G**
Issued:	**12/08/2010**
Committee:	**SC-135**
Description:	DO-160G provides standard procedures and environmental test criteria for testing airborne equipment for the entire spectrum of aircraft from light general aviation aircraft and helicopters through the "jumbo jets" and SST categories of aircraft. The document includes 26 Sections and three Appendices. Examples of tests covered include vibration, power input, radio frequency susceptibility, lightning and electrostatic discharge. **Coordinated with EUROCAE, RTCA/DO-160G and EUROCAE/ED-14G are identically worded.** DO-160G is recognized by the International Organization for Standardization (ISO) as de facto international standard ISO- 7137.

Questions?

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